TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	65.7 ± 2.83
Correlation Stability	88.1 ± 0.70
Distribution Consistency	81.0 ± 6.51
Entropy Preservation	29.2 ± 1.83
Mean Z Score Drift	0.4 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	6.0 ± 1.00
Data Density	100.0 ± 0.00
Coverage Score	87.1 ± 1.70
Feature Accuracy	63.7 ± 1.00
Novelty Score	0.7 ± 0.02

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	56.285	+1.846	3.4%	OK
sex	0.680	0.649	-0.031	4.6%	OK
cp	3.158	3.568	+0.409	13.0%	Warn
trestbps	131.690	136.167	+4.477	3.4%	OK
chol	246.693	250.695	+4.002	1.6%	OK
fbs	0.149	0.037	-0.112	75.2%	Drift
restecg	0.990	0.998	+0.007	0.8%	OK
thalach	149.607	137.032	-12.575	8.4%	OK
exang	0.327	0.302	-0.025	7.5%	OK
oldpeak	1.040	1.749	+0.709	68.2%	Drift
slope	1.601	1.703	+0.102	6.4%	OK
ca	0.672	1.321	+0.649	96.6%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	65.7 ± 2.83
Correlation Stability (*2)	88.1 ± 0.70
Distribution Consistency (*3)	81.0 ± 6.51
Entropy Preservation (*4)	29.2 ± 1.83
Mean Z Score Drift (*5)	0.4 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	6.0 ± 1.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	87.1 ± 1.70
Feature Accuracy (*10)	63.7 ± 1.00
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.15081967213114753	0.5327868852459018	-0.014754098360655759	
F1	0.534375869626353	0.09870500963995517	0.5399339734913983	0.005558103865045211	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.15409836065573773	0.5557377049180328	0.013114754098360654	
F1	0.5198013329305267	0.10049297582897307	0.5536880134200428	0.03388668048951599	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5278688524590164	0.12622950819672132	0.4721311475409836	-0.0557377049180328	*
F1	0.5325413347450619	0.07997341100583041	0.4966327320991934	-0.0359086026458625	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.14098360655737704	0.5098360655737705	-0.04754098360655737	*
F1	0.568660979990255	0.07787560508731783	0.5431842103058855	-0.025476769684369537	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.05573770491803278	0.4196721311475410	-0.1278688524590164	*
F1	0.3899270600110827	0.03591494075281135	0.4505319761287252	0.060604916117642495	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.16557377049180327	0.5672131147540983	-0.04098360655737705	
F1	0.6070236204175147	0.10300906943285715	0.5781532265241386	-0.028870393893376013	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5868852459016394	0.16721311475409834	0.5573770491803278	-0.02950819672131147	
F1	0.54293746263323	0.1160435430765567	0.552491870078987	0.009554407445757163	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.15901639344262294	0.5081967213114753	-0.09016393442622954	*
F1	0.5648347697855012	0.10779426608057738	0.5348285755222246	-0.030006194263276505	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5180327868852459	0.11803278688524589	0.4655737704918026	-0.05245901639344262	
F1	0.4850631713299050 6	0.07909880567048906	0.4877986056378417	0.002735434307936663	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.09672131147540983	0.3098360655737705	-0.24426229508196723	*
F1	0.4442202071436568 4	0.06831741461423718	0.3623032914423190 4	-0.08191691570133781	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	57.0	0.0	4.0	150.0	239.0	0.0	0.0	159.0	0.0	1.1140344107424618
2	42.0	1.0	3.0	148.0	225.0	0.0	0.0	175.0	0.0	0.17419966666245396
3	64.0	1.0	3.0	134.0	326.0	0.0	0.0	148.0	0.0	0.6192669099170631
4	62.0	1.0	3.0	125.0	306.0	0.0	0.0	142.0	0.0	1.2606898035473553
5	59.0	0.0	3.0	159.0	302.0	0.0	2.0	160.0	0.0	0.028895675684565747
6	61.0	0.0	3.0	150.0	317.0	0.0	1.0	159.0	0.0	0.0
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	51.0	1.0	2.0	117.0	229.0	0.0	0.0	160.0	0.0	0.7564465776716754
2	50.0	1.0	3.0	114.0	238.0	0.0	0.0	151.0	0.0	0.1276746368824741
3	53.0	1.0	3.0	117.0	229.0	0.0	0.0	156.0	0.0	0.14702726999913057
4	59.0	0.0	1.0	168.0	289.0	0.0	2.0	158.0	0.0	0.2490380083663459 3
5	59.0	0.0	1.0	163.0	292.0	0.0	2.0	155.0	0.0	0.4103043258072152
6	60.0	0.0	4.0	154.0	305.0	0.0	2.0	157.0	0.0	0.2551444899352375 3
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	48.0	1.0	4.0	129.0	250.0	0.0	2.0	142.0	1.0	1.0070603858410414
2	54.0	1.0	3.0	124.0	265.0	0.0	0.0	160.0	0.0	0.6823876254020464
3	58.0	1.0	2.0	122.0	271.0	0.0	2.0	151.0	0.0	2.2620441456759908
4	63.0	0.0	3.0	137.0	333.0	0.0	0.0	160.0	0.0	0.0
5	60.0	0.0	3.0	129.0	326.0	0.0	2.0	168.0	0.0	0.13342211942000226
6	51.0	1.0	3.0	125.0	253.0	0.0	0.0	160.0	0.0	0.0
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	54.0	1.0	1.0	190.0	283.0	0.0	2.0	192.0	0.0	0.014147587979604227
2	58.0	0.0	3.0	175.0	251.0	0.0	0.0	146.0	0.0	0.0
3	59.0	1.0	3.0	140.0	185.0	0.0	2.0	155.0	0.0	3.005527664980239
4	61.0	1.0	3.0	147.0	208.0	0.0	2.0	151.0	0.0	1.6382893896908586
5	56.0	1.0	4.0	132.0	185.0	0.0	2.0	107.0	1.0	2.0826232442747778
6	56.0	1.0	4.0	131.0	191.0	0.0	2.0	111.0	1.0	2.123599336722621
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	55.0	1.0	2.0	137.0	203.0	0.0	1.0	157.0	0.0	2.211495324190186
2	53.0	1.0	3.0	139.0	203.0	0.0	1.0	155.0	0.0	2.9051392903659092
3	45.0	1.0	3.0	123.0	282.0	0.0	2.0	157.0	0.0	0.7884014812023326
4	58.0	1.0	3.0	123.0	294.0	0.0	2.0	167.0	0.0	0.6334709786055264
5	57.0	1.0	3.0	128.0	224.0	0.0	2.0	143.0	0.0	1.0085188987936482
6	58.0	1.0	4.0	130.0	214.0	0.0	2.0	132.0	1.0	2.1432271572111885
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	60.0	0.0	3.0	148.0	304.0	0.0	2.0	147.0	0.0	0.8830956705967512
2	59.0	1.0	1.0	160.0	274.0	0.0	2.0	127.0	0.0	0.016463291783600426
3	44.0	1.0	4.0	131.0	247.0	1.0	2.0	143.0	1.0	0.2113731568072146 6
4	57.0	1.0	4.0	142.0	270.0	0.0	2.0	120.0	1.0	1.2250767558933 38
5	60.0	1.0	4.0	127.0	269.0	0.0	0.0	102.0	1.0	1.5051868420848638
6	60.0	1.0	4.0	126.0	256.0	0.0	1.0	142.0	1.0	2.3288389533675
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	52.0	1.0	4.0	127.0	254.0	0.0	0.0	160.0	0.0	0.006221572770353068
2	60.0	1.0	3.0	109.0	246.0	0.0	1.0	156.0	0.0	0.43719647186345195
3	50.0	1.0	2.0	119.0	199.0	0.0	1.0	170.0	0.0	0.32455598780442885
4	57.0	1.0	2.0	130.0	217.0	0.0	1.0	155.0	0.0	0.5908972169806704
5	53.0	1.0	1.0	138.0	209.0	0.0	0.0	167.0	0.0	0.5874568923690043
6	51.0	1.0	2.0	140.0	211.0	0.0	0.0	169.0	0.0	0.5021790396277784
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	40.0	1.0	3.0	124.0	282.0	0.0	2.0	156.0	0.0	0.3987293482324042
2	47.0	1.0	2.0	122.0	283.0	0.0	2.0	158.0	0.0	1.0000530060829014
3	35.0	1.0	4.0	125.0	282.0	0.0	2.0	156.0	0.0	0.0
4	62.0	0.0	3.0	125.0	196.0	0.0	0.0	137.0	0.0	0.19109164020065172
5	59.0	1.0	3.0	135.0	196.0	0.0	2.0	146.0	0.0	2.7061826894262728
6	61.0	1.0	3.0	151.0	216.0	0.0	1.0	149.0	0.0	1.5141030305943
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	64.0	1.0	3.0	125.0	309.0	0.0	0.0	129.0	0.0	1.845575535330636
2	63.0	0.0	3.0	125.0	308.0	0.0	0.0	131.0	1.0	1.7808090895390551
3	57.0	0.0	4.0	139.0	241.0	0.0	0.0	124.0	1.0	0.28406110539808926
4	60.0	0.0	4.0	133.0	237.0	0.0	0.0	124.0	1.0	0.9255730438389427
5	56.0	1.0	4.0	126.0	255.0	0.0	2.0	141.0	1.0	2.1901278204565
6	53.0	0.0	3.0	119.0	164.0	0.0	1.0	129.0	0.0	0.535719294025341
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	4.0	130.0	217.0	0.0	2.0	133.0	1.0	2.2384365766267025
2	60.0	1.0	3.0	140.0	186.0	0.0	2.0	153.0	0.0	2.934324234776507
3	63.0	1.0	4.0	131.0	216.0	0.0	2.0	134.0	1.0	2.192251199623922
4	60.0	0.0	4.0	132.0	236.0	0.0	0.0	125.0	1.0	1.1054722601847005
5	56.0	0.0	4.0	126.0	248.0	0.0	1.0	141.0	1.0	1.0825055211361183
6	57.0	0.0	4.0	144.0	224.0	0.0	0.0	110.0	0.0	1.4813226383247604
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6